

University of Bahrain
Collage of Engineering
Department of Chemical engineering
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PICENG 290

Arduino Weather Station

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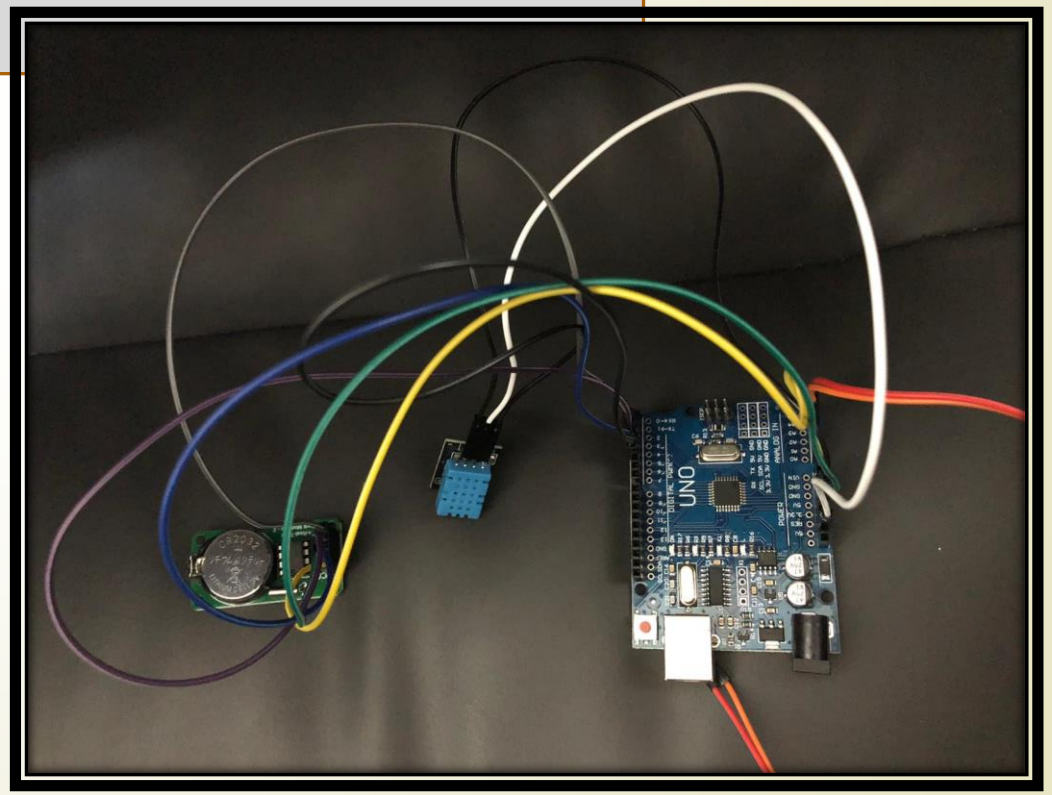
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Introduction

- A Weather station is a device that collects data related to the weather and environment using many different sensors. We can measure many things like :
- Temperature
- Humidity
- Wind
- Barometric Pressure
- UV index
- Rain

In the Weather Station that we are going to build we are going to measure Temperature and Humidity in Different locations and display the current date and time.

- Building a Weather Station is extremely easy and it means that you don't need anymore inaccurate results from your weather channel ! You can log the data and also play around with it.



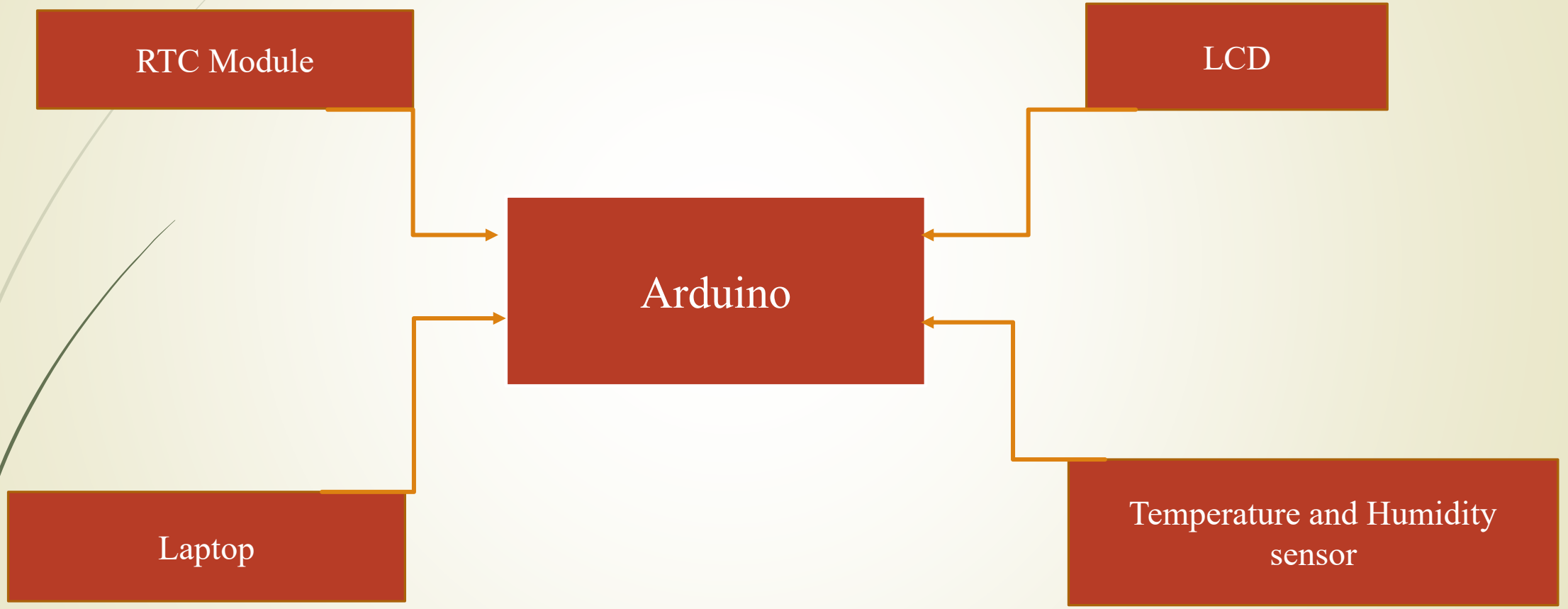
Working principle

- The device works by taking readings from various sensors at different pins in Arduino microcontroller. All sensors are attached to the microcontroller each of them taking 5v input from the Arduino.
- For the temperature sensor to prevent any damage or unstable behavior a 10kohm resistor is already attached to it with a Decoupling capacitor to filter out the noise on the power supply. It gives us continuous reading of the surrounding environment in the range of 2 to 3 seconds.
- All of the components are connected to the Arduino by codes typed and run on the Arduino Sketch software.

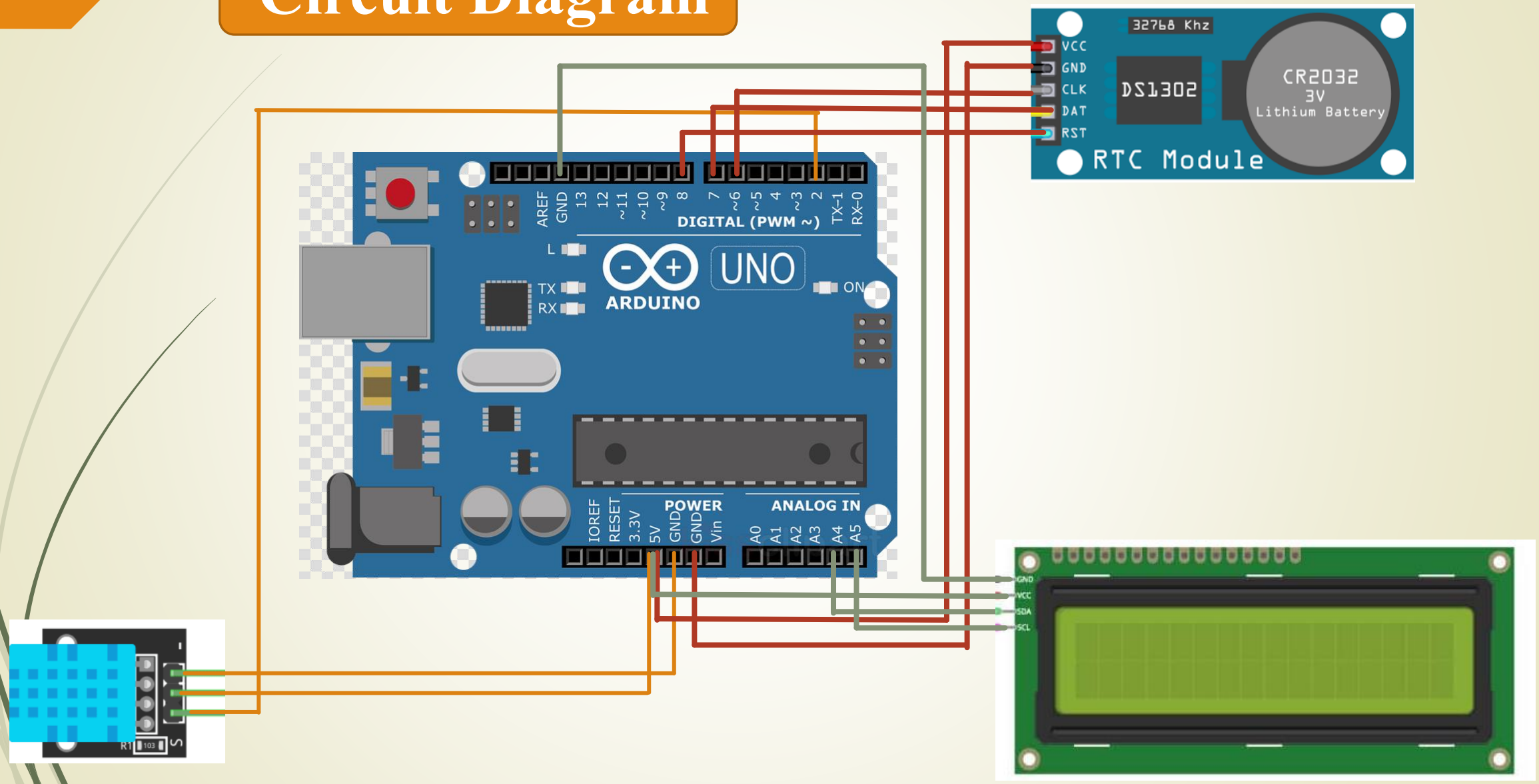
Advantages

- The Arduino Weather station has the capability of working on low power. Hence it is not much dependent on power source.
- The components are widely available and inexpensive.
- It also has around ± 1 unit error accuracy.

Block Diagram



Circuit Diagram

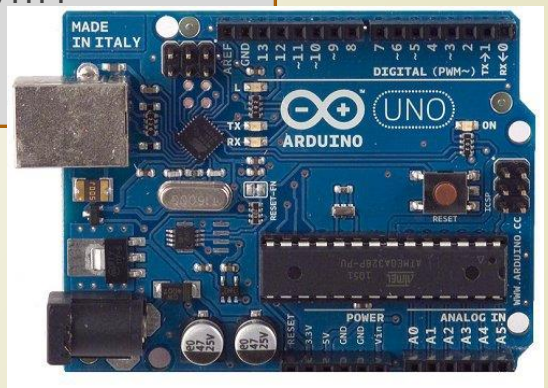


Hardware Components

- Arduino Uno
- LCD (16X2)
- RTC Module (DS3231)
- DHT11 Temperature and Humidity sensor
- Bread Board

What is Arduino?

- Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language ,and the Arduino Software (IDE), based on Processing.
- Thanks to its simple and accessible user experience, Arduino has been used in thousands of different projects and applications. The Arduino software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows, and Linux. Teachers and students use it to build low cost scientific instruments, to prove chemistry and physics principles, or to get started with programming and robotics.

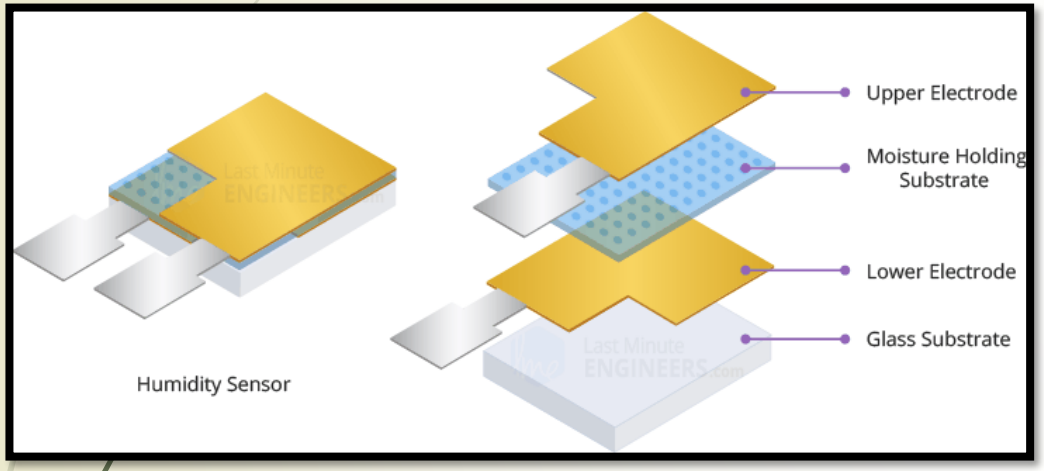


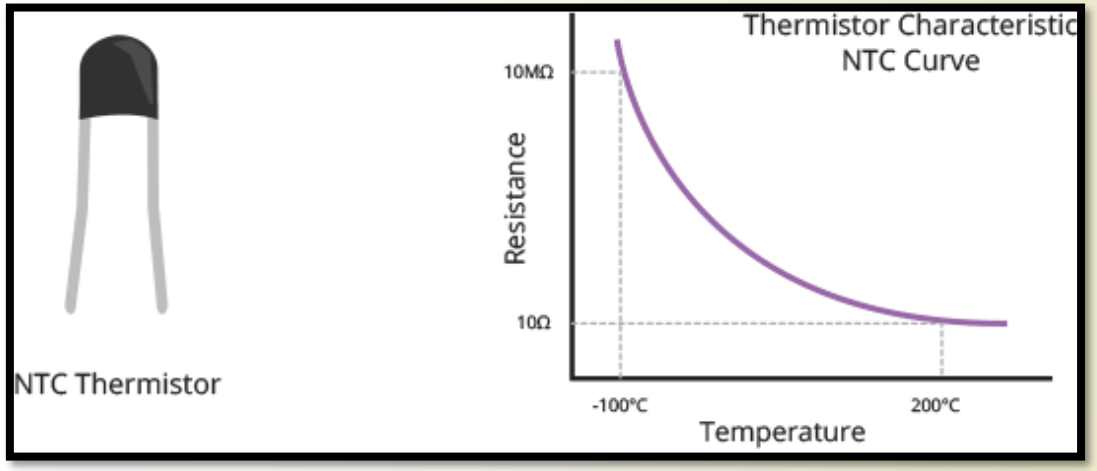
What is a Sensor ?

- A sensor is a device that detects and responds to some type of input from the physical environment. The specific input could be light, heat, motion, moisture, pressure, or any one of a great number of other environmental phenomena. The output is generally a signal that is converted to human-readable display at the sensor location or transmitted electronically over a network for reading or further processing.

DHT11 Sensor

- Inside the DHT11, there is a Humidity sensing component and a Thermistor.

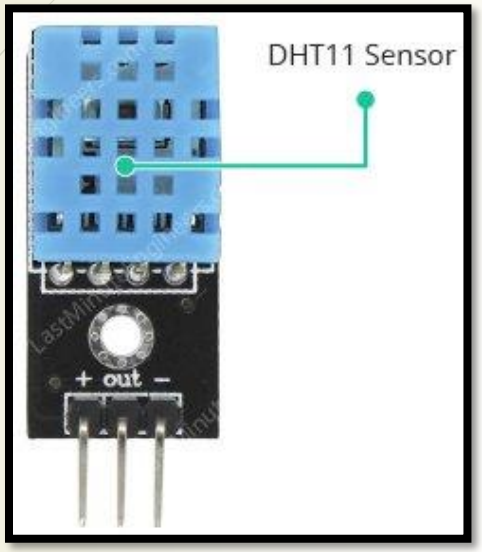


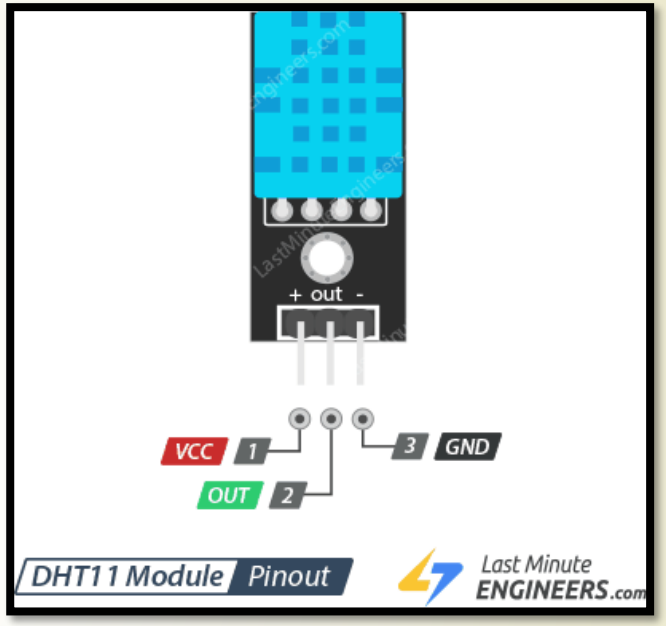


The Humidity sensing component has two electrodes with moisture holding substrate sandwiched between them. The change in resistance is proportional to the relative humidity between the electrodes.

DHt11 also contains a NTC/Thermistor to measure temperature. The term "NTC" means "Negative Temperature Coefficient", which means that the resistance decreases with increase of the temperature.

Pinout and specification





DHT11 can measure

- Temperature from 0°C to 50°C with ±2.0°C accuracy,
- Humidity from 20 to 80% with 5% accuracy.
- The sampling rate of the DHT11 is 1Hz, meaning you can get new data from it once every second.

- VCC- supplies power for the sensor. Range 3.3v to 5v.
- Out- used for communication btw Arduino and sensor.(Digital pin#8).
- GND- connected to ground of the Arduino.

Advantage

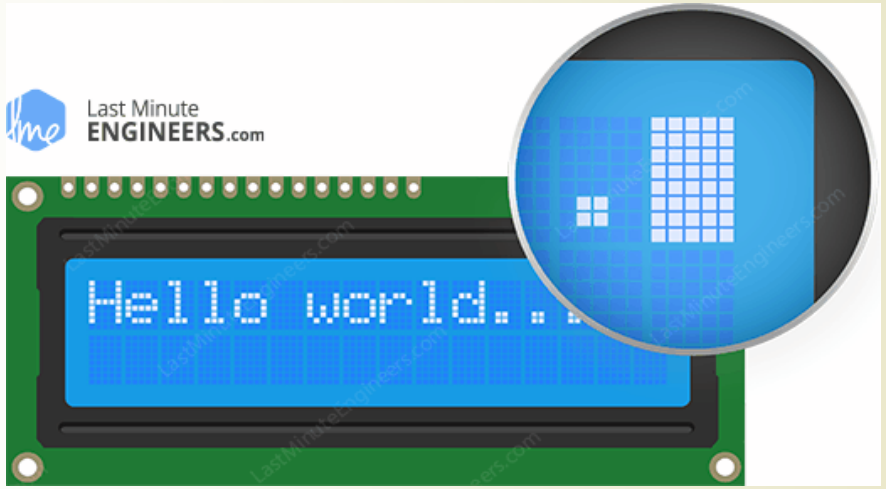
- Ultra low cost.
- Smaller size and
- Has higher sampling rate,ie,1Hz.

Disadvantage

In this sensor you can get new data from it once every one second or two.

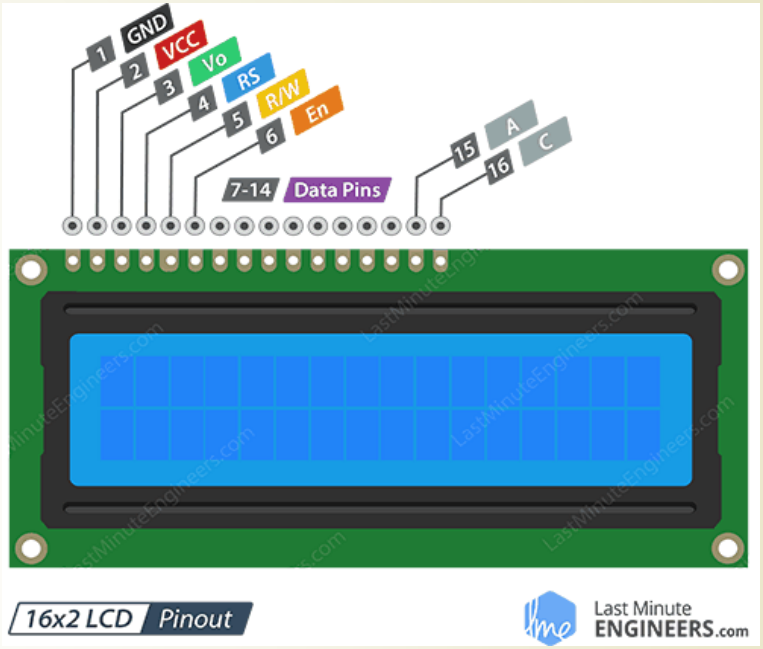
16x2 LCD- Liquid Crystal Display

- It is basically a display unit which uses liquid crystals to produce a visible image.
- When current is applied to this special kind of crystal, it turns opaque blocking the backlight that lives behind the screen. As a result that particular area will become dark compared to other. And that's how characters are displayed on the screen.
- The display has an LED backlight and can display 32 ASCII characters in two rows with 16 characters on each row.



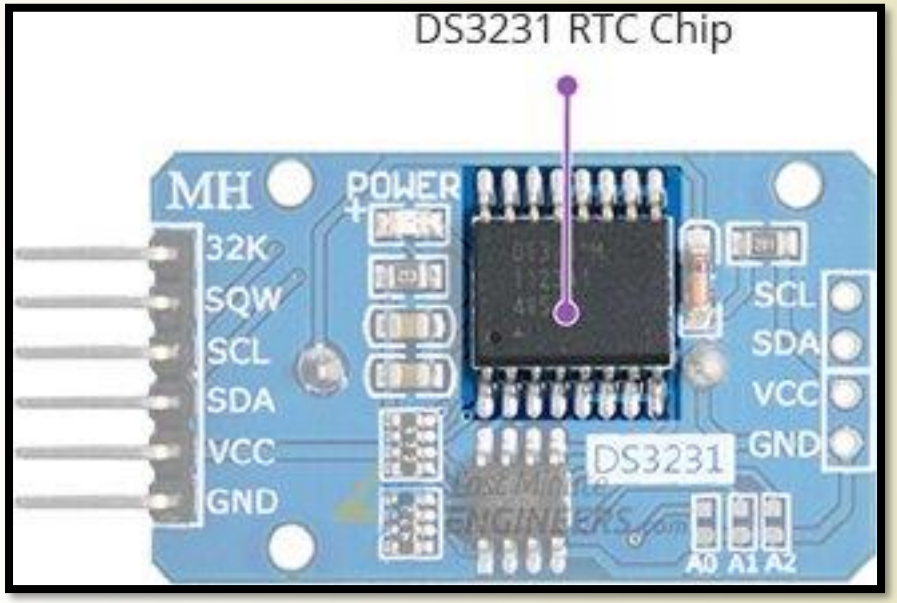
Pinout

- GND- should be connected to the ground of the Arduino.
- VCC- is the power supply for the LCD which we connect the 5 volts pin on the Arduino.
- Vo- controls the contrast and brightness of the LCD
- RS(Register Select)- pin lets the Arduino tell the LCD whether it is sending commands or the data. Basically this pin is used to differentiate commands from the data.
- R/W(Read/Write)- pin on the LCD is to control whether or not you're reading data from the LCD or writing data to the LCD. Since we're just using this LCD as an OUTPUT device, we're going to tie this pin LOW. This forces it into the WRITE mode.
- EN(Enable)- enable the display.
- Data pins- carries the 8 bit data we send to the display.
- A-C(Anode&Cathode)- controls the backlight of the LCD.



RTC- Real Time Clock

- The module is a low-cost, extremely accurate RTC chip from Maxim – DS3231.
- It manages all timekeeping functions and features a simple two-wire I2C interface which can be easily interfaced with any microcontroller of your choice.
- The chip maintains seconds, minutes, hours, day, date, month, and year information.
- The clock operates in either the 24-hour or 12-hour format with an AM/PM indicator. It also provides two programmable time-of-day alarms



Battery Backup

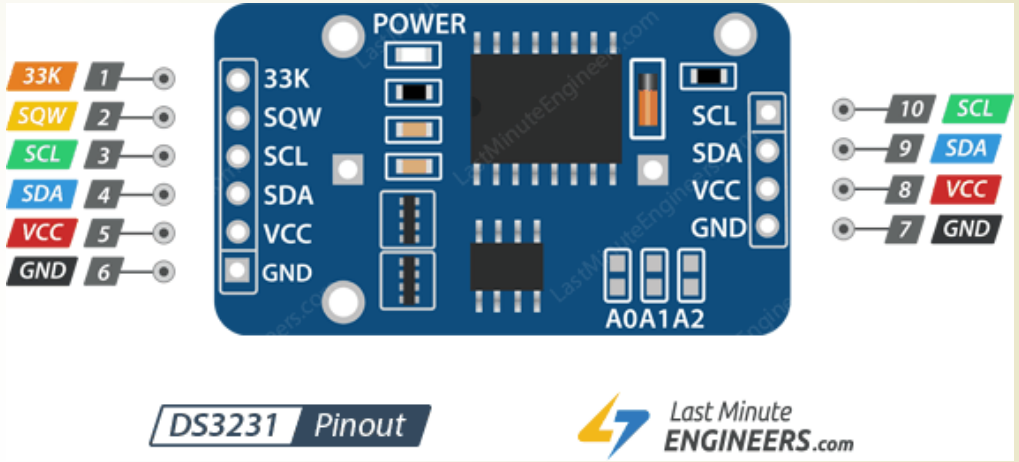
- DS3231 has a battery backup which maintains accurate time keeping when the main power of the device is interrupted.
- It monitors the status of VCC to detect power failure and automatically switches to backup supply.
- Bottom side hold a battery holder for 20mm,3v lithium coin cells. Any CR2032 battery will fit well.

Without RTC connected to 5v supply it can run for a minimum of 8 years.



Pin out

- 33k- pin outputs the stable(temperature compensated) and accurate reference clock.
- SQW- pin outputs a nice square wave at either 1Hz, 4kHz, 8kHz or 32kHz and can be handled programmatically. This can further be used as an interrupt due to alarm condition in many time-based applications.
- SCL- is a serial clock pin for I2C interface
- SDA- is a serial data pin for I2C interface.
- VCC- pin supplies power for the module. It can be anywhere between 3.3V to 5.5V.
- GND- is a ground pin.

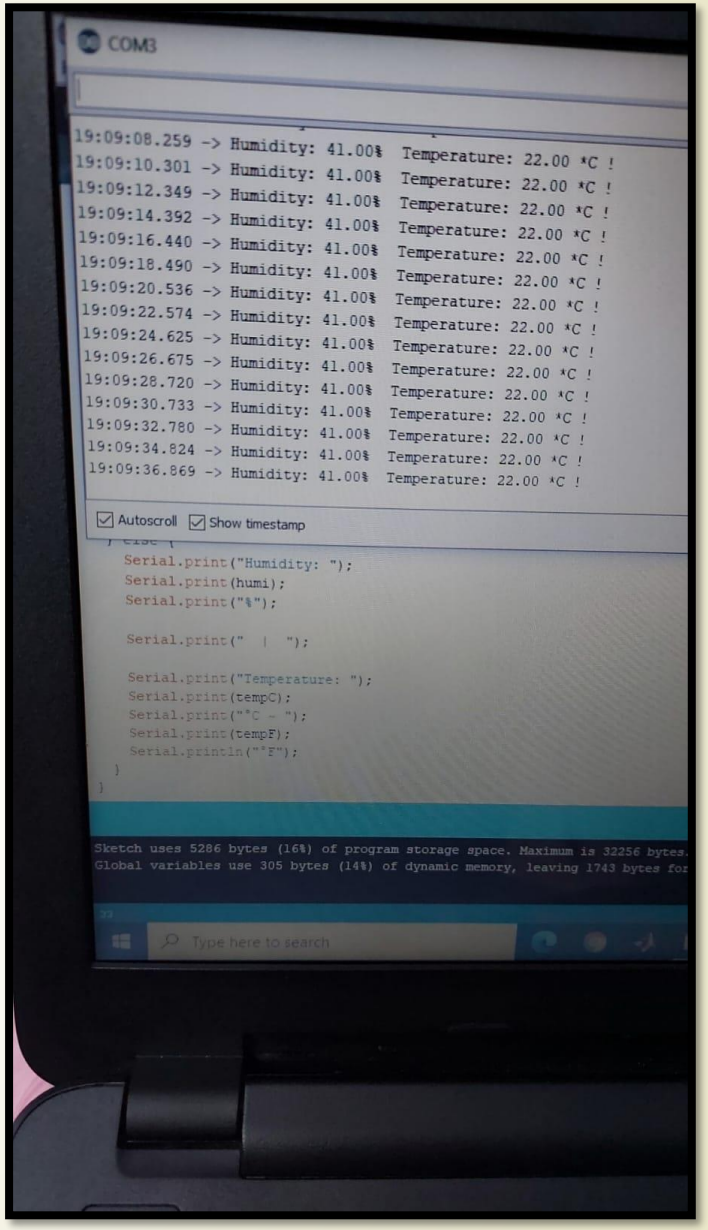


Connections with Arduino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <virtuabotixRTC.h>
#include "DHT.h"
#define DHTPIN 2
#define DHTTYPE DHT11


LiquidCrystal_I2C lcd(0x27, 16, 2);  // I2C address 0x3F, 16 column and 2 rows
DHT dht(DHTPIN, DHTTYPE);
virtuabotixRTC myRTC(6, 7, 8);


void setup()
{
   Serial.begin(9600);
// Set the current date, and time in the following format:
// seconds, minutes, hours, day of the week, day of the month, month, year
  myRTC.setDS1302Time(00, 29, 8, 4, 18, 11, 2020);
  dht.begin();     // initialize the sensor
  lcd.init();      // initialize the lcd
  lcd.backlight(); // open the backlight
}
```



```
void loop()
{
  delay(2000); // wait a few seconds between measurements


 float humi  = dht.readHumidity();    // read humidity
  float tempC = dht.readTemperature(); // read temperature


   lcd.setCursor(0, 1);  // start to print at the second row
    lcd.print(tempC);     // print the temperature
    lcd.print((char)223); // print ° character
    lcd.print("C ");
    lcd.print(humi);      // print the humidity
    lcd.print("%");
    // This allows for the update of variables for time or accessing the individual elements.
  myRTC.updateTime();

 lcd.setCursor(0,0);   // print at the first row
  lcd.print(myRTC.dayofmonth);

  lcd.print(myRTC.year);
  lcd.print(" ");
  lcd.print(myRTC.hours);
  lcd.print(":");

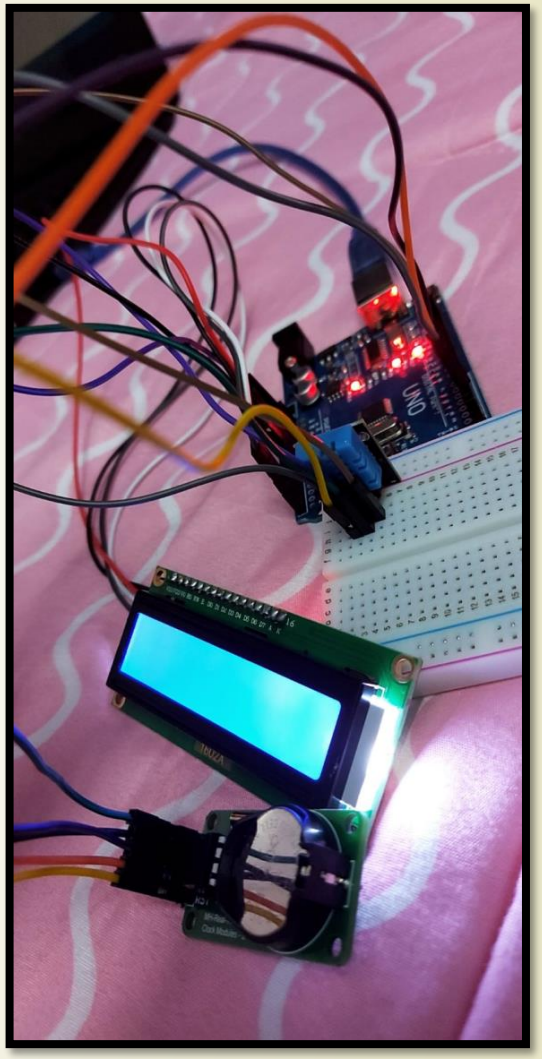
 lcd.println(myRTC.seconds);
delay( 2000);
   Serial.print(myRTC.dayofmonth);

  Serial.print("/");
  Serial.print(myRTC.month);
  Serial.print("/");
  Serial.print(myRTC.year);
  Serial.print("  ");
  Serial.print(myRTC.hours);
  Serial.print(":");
  Serial.print(myRTC.minutes);
  Serial.print(":");
  Serial.println(myRTC.seconds);
```

```
  Serial.print("Humi:");
  Serial.print(humi);
  Serial.print("%");
  Serial.print("Temp");
  Serial.print(tempC);
  Serial.print("C");
                                                        //|

// Delay so the program doesn't print non-
stop
  delay( 2000);


}
```



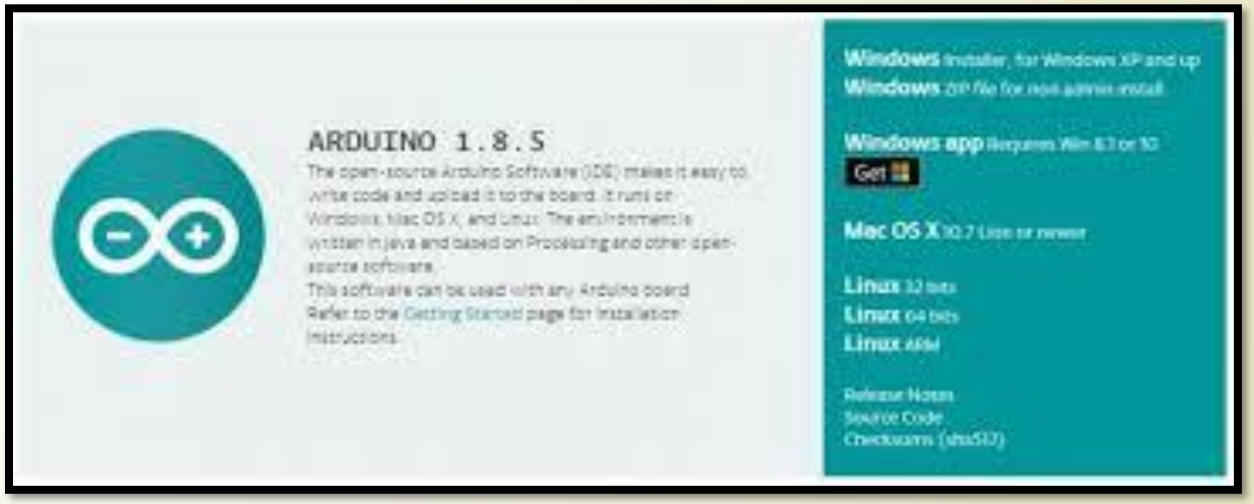
Software and Library

- The software that is used to program the microcontroller, is open-source-software and can be downloaded for free on www.arduino.cc. With this "Arduino software" you can write little programs witch the microcontroller should perform. This programs are called "Sketch".

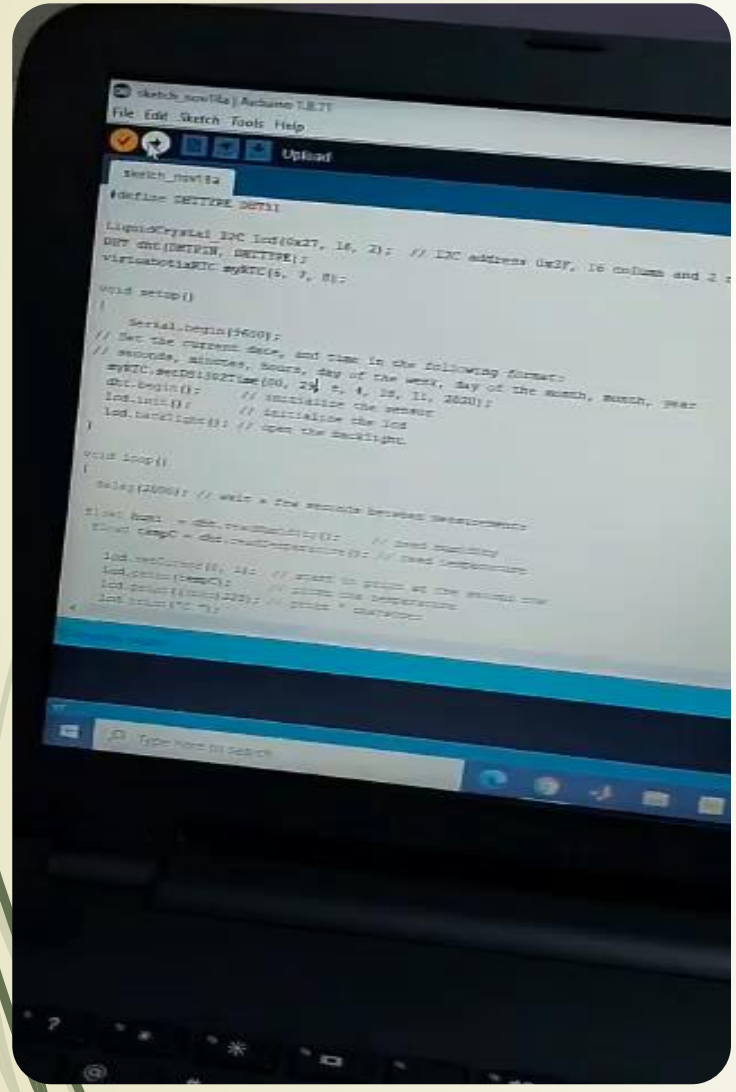
  In the end the sketches are transferred to the microcontroller by USB cable

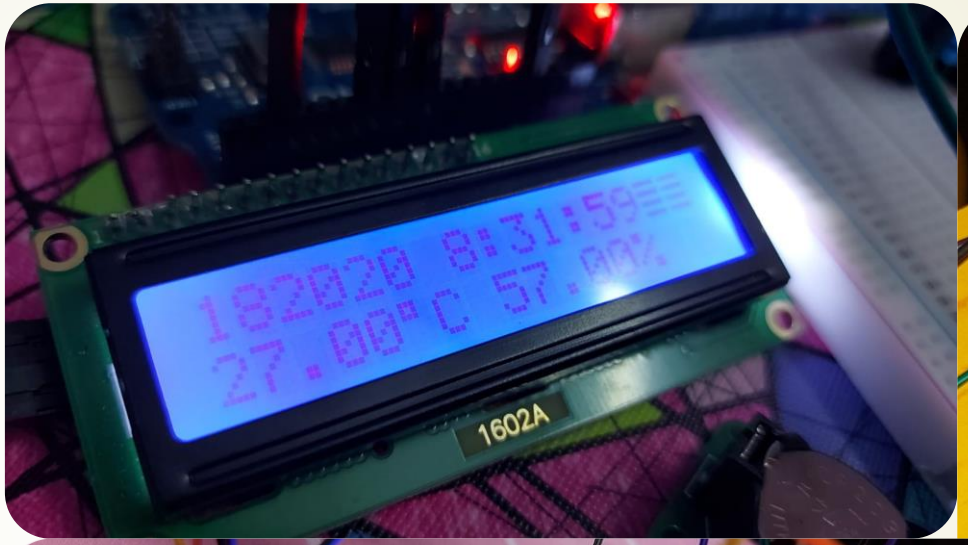
Some of libraries that we used :

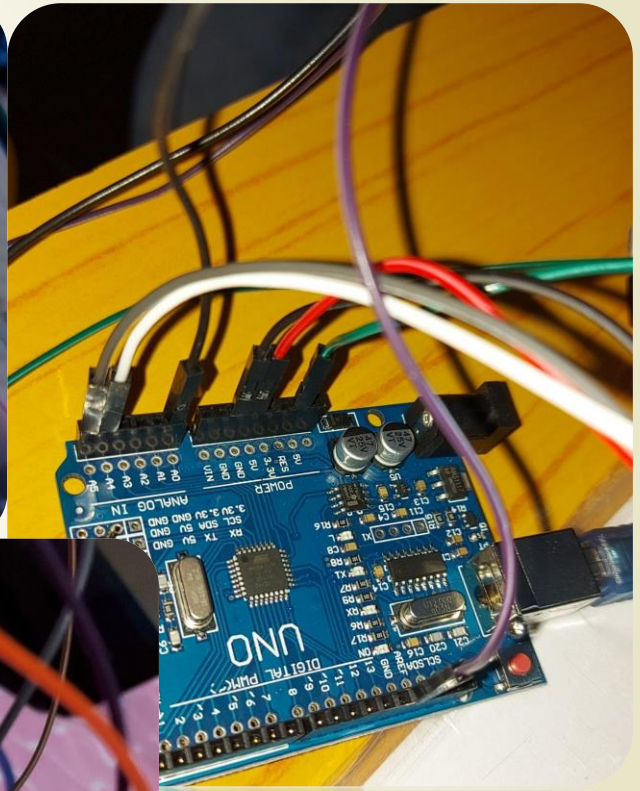
- LCD library : <LiquidCrystal.h>
- RTC library : "RTClib.h"
- DHT11 library : <dht.h>

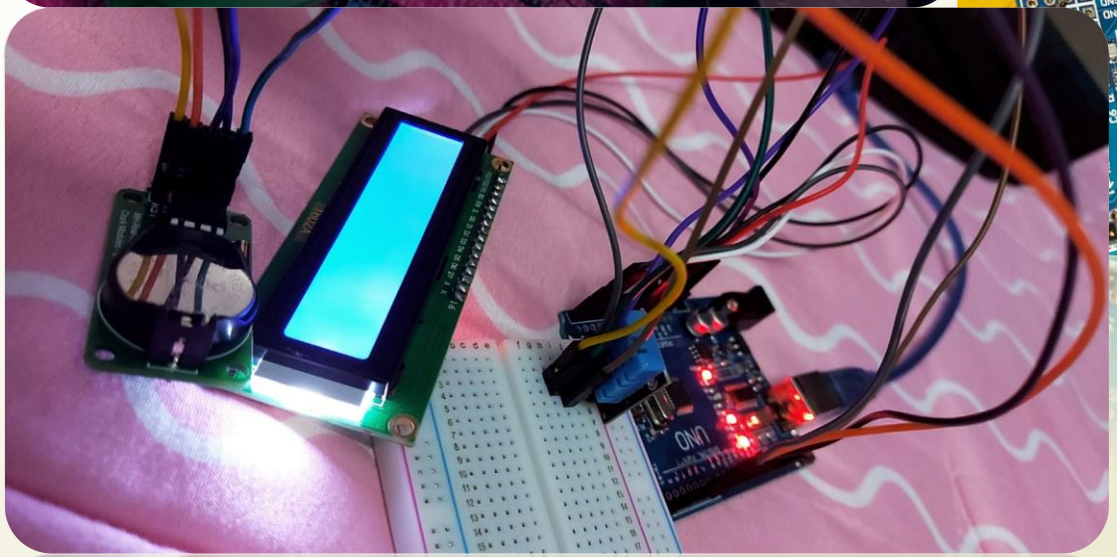


Project progress and Video









Applications

- As an application, a normal person can place this device at various places like in his backyard garden for soil moisture and rain water reading, indoor swimming pool for the maintance of water temperature and humidity in the air.

- This weather station has an external feature of website access all around the world. It attempts to show live feed of readings from that environment where the results are required.

Future Vision

- Here our project is a simple Weather station that measures Temperature and Humidity by using a DHT11 sensor.
- For our future vision of this project we would like to work with many different sensors to help us read other climatic factors like Rain,Pressure,UV index etc.
- In further expansion of this simple small scale project, we can make it in large scale by building up a system that has Wireless connectivity .To do this we can connect a CC3000 Wi-Fi shield over the Arduino to connect it to local internet connection providers and connect. Its job is to transmit the data to a website linked to it and visualize the data over there for every 1minute or 30 seconds.
- Also to make it compatible with smartphones apps to give any critical feedback of data. Updating twitter status and performing actions on the basis of that which is one of the most efficient use of IOT(Internet of things).The special feature to be included as an idea in this device is that it can be used for any critical environments or local area rather than an expensive weather station that is capable of performing over a large scale. These work on small scale too on public wireless LANs.

Conclusion

- Arduino Weather station is a easy and simple project to implement. In this project we built the weather station to help us measure temperature and humidity in a given location. The hardware components used were easily available and at a low cost.
- The components were carefully connect to the Arduino Uno and sketched. The data analysis we received were satisfying.
- Most importantly, it is fun having a small and handy weather station in your homes or anywhere you want it.
- In this project we learned how the sensors work and how we can broaden our views to make much more interesting project in the future by just using Arduino .

Thankyou for Listening .